INTEGRAL POR SUBSTITUIÇÃO: *Inverso da regra da cadeia

$$\int (x+1)^{17} dx = \frac{(x+1)^{18}}{18}$$

$u = (x+1)$ $\quad \frac{du}{dx} = 1$ $\quad du = dx$

$$\boxed{\int a x^n dx = \frac{a \cdot x^{n+1}}{n+1} + C}$$

$$\int u^{17} du = \frac{u^{18}}{18} + C$$

$$\int e^{\operatorname{sen} x} \cdot \cos x \, dx$$

$u = \operatorname{sen} x$ $\quad \frac{du}{dx} = \cos u$

$du = \cos u \cdot dx$

$$\int e^u du = e^u + C = \boxed{e^{\operatorname{sen} x} + C}$$

$$\int \frac{\operatorname{sen}\sqrt{x}}{\sqrt{x}} dx$$

$u = \sqrt{x}$ $\quad \frac{du}{dx} = \frac{1}{2} \cdot x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$ $\quad 2du = \frac{dx}{\sqrt{x}}$

$u = x^{\frac{1}{2}}$

$$\int \operatorname{sen}\sqrt{x} \cdot \frac{du}{\sqrt{x}} = \int \operatorname{sen} u (2du) = 2 \cdot (-\cos u) + C = \boxed{-2\cos u + C}$$

$$\int \frac{\cos(\ln(x))}{x} dx \quad u = \ln x \quad \frac{du}{dx} = \frac{1}{x} \quad dx = du \, x$$

$$\int \frac{\cos(u)}{\cancel{x}} du \, \cancel{x} = \operatorname{Sen} u + C = \operatorname{Sen}(\ln x) + C$$

$$\int \frac{\operatorname{sen} x}{\sqrt{1-\cos x}} dx \quad \boxed{u = 1 - \cos x} \quad \boxed{\frac{du}{dx} = \operatorname{sen} x}$$

$$\boxed{du = dx \cdot \operatorname{sen} x}$$

$$\int \frac{du}{\sqrt{u}} = \int u^{-\frac{1}{2}} du = \frac{u^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} + C = \boxed{2\sqrt{1-\cos x} + C}$$

$$\int \cos(x) \cdot \operatorname{Sen}^5(x) \cdot dx \quad \boxed{u = \operatorname{sen}(x)} \quad \boxed{\frac{du}{dx} = \cos(x)}$$

$$\boxed{du = \cos(x) \cdot dx}$$

$$\int u^5 du = \frac{\operatorname{Sen}^6(x)}{6} + C$$

$$\int \sqrt{5x-1} \, dx \quad u = 5x-1 \quad \frac{du}{dx} = 5 \quad dx = \frac{du}{5}$$

$$\int \sqrt{u} \cdot \frac{du}{5} = \frac{1}{5}\int u^{\frac{1}{2}} du = \frac{1}{5} \cdot \frac{u^{\frac{1}{2}+1}}{\frac{1}{2}+1} + C = \frac{1}{5} \cdot \frac{u^{\frac{3}{2}}}{\frac{3}{2}} = \boxed{\frac{2}{15} \cdot (5x-1)^{\frac{3}{2}}}$$

INTEGRAL POR PARTES:

*Inverso da derivada do produto

$$\boxed{\frac{d}{dx}(f(x) \cdot g(x)) = f'(x) \cdot g(x) + f(x) \cdot g'(x)}$$

Integrando os dois lados:

$$\int (f(x) \cdot g(x))' = \int f(x)' \cdot g(x) + \int f(x) \cdot g(x)' \Longrightarrow f(x) \cdot g(x) = \int f(x)' \cdot g(x) + \int f(x) \cdot g(x)' \Longrightarrow$$

$$\Longrightarrow \boxed{\int f(x)' \cdot g(x) = f(x) \cdot g(x) - \int f(x) \cdot g(x)'}$$

ou em outras variáveis: $\boxed{\int u \cdot dv = u \cdot v - \int v \cdot du}$

*Variável solto você deriva

*Derivado você integra.

$\oplus$ $\ominus$

LIATE → logarítmico, trig inverso, algébrico, trig, exponencial

$$\int \underbrace{x}_{\text{DERIVAR: } f(x)} . \underbrace{\text{sen}(x)}_{\text{INTEGRAR: } g'(x)} \, dx =$$

$f(x) = x \qquad g'(x) = \text{sen}\,x$

$f'(x) = 1 \qquad g(x) = -\cos x$

$$\overset{*}{=} -x.\cos x - \int 1.(-\cos x)\, dx =$$

$$= -x.\cos(x) + \int \cos x \, dx = \boxed{-x.\cos x + \text{sen}\,x + C}$$

$$\int \underbrace{(2x-1)}_{\text{DERIVAR: } f(x)} . \underbrace{e^x}_{\text{INTEGRAR: } g'(x)} \, dx =$$

$f(x) = 2x-1 \qquad g'(x) = e^x$

$f'(x) = 2 \qquad g(x) = e^x$

$$= (2x-1).e^x - \int 2.e^x \, dx = (2x-1).e^x - 2e^x + C =$$

$$= (2x-3).e^x + C$$

$$\int x^3 . e^{x^2} . dx = \int x^2 . \overbrace{e^{x^2} . x . dx}^{u . e^u . x . dx} \overset{*}{=}$$

$u = x^2 \qquad \frac{du}{dx} = 2x \qquad \boxed{x.dx = \frac{du}{2}}$

$$\overset{*}{=} \int u . e^u \frac{du}{2} = \frac{1}{2} \int \underbrace{u}_{\text{DERIVAR}} . \underbrace{e^u}_{\text{INTEGRAR}} = \frac{1}{2} u . e^u - \int 1 . e^u = \frac{1}{2}(u.e^u - e^u) = \frac{(e^u . (u-1))}{2} \overset{*}{=} \boxed{\frac{e^{x^2} . (x^2 - 1)}{2}}$$

$f(x) = u \qquad g'(x) = e^u$

$f'(x) = 1 \qquad g(x) = e^u$

INTEGRAL POR COMPLETAMENTO DE QUADRADOS:

$x^2 - 6x + 5 = 0 \quad (a-b)^2 = a^2 - 2ab + b^2) \qquad \left(x - \frac{6}{2}\right)^2 + 5 - \left(\frac{6}{2}\right)^2 = 0 \qquad (x-3)^2 = 4 \qquad x = 5 \quad x = 1$

*Polinomio no denominador tem raízes complexas. $\boxed{\Delta < 0}$

$$\int \frac{dx}{x^2 + 2x + 2} = \int \frac{dx}{(x+1)^2 + 1} = \int \frac{du}{u^2 + 1} = \text{arctg}\, u + C \overset{*}{=} \boxed{\text{arctg}(x+1) + C}$$

$$\boxed{\text{Sabemos que } \frac{d(\text{arctg}\, x)}{dx} = \frac{1}{x^2+1} \text{ e } \int \frac{1}{x^2+1} dx = \text{arctg}\, x + C}$$

$x^2 + 2x + 2 = (x+1)^2 - 1 + 2 = (x+1)^2 + 1$

$u = x + 1 \quad \therefore \quad \frac{dx}{du} = 1 \Rightarrow dx = du$

$$\int \frac{1}{x^2 + 8x + 52} = \int \frac{dx}{(x+4)^2 + 36} = \int \frac{dx}{36 . \left(\frac{(x+4)^2}{36} + 1\right)} = \frac{1}{36} \int \frac{dx}{\left(\frac{x+4}{6}\right)^2 + 1} \overset{*}{=}$$

$u = \left(\frac{x+4}{6}\right)$

$\frac{du}{dx} = \frac{1}{6} \qquad 6du = dx$

$$\overset{*}{=} \frac{1}{36} \int \frac{6\, du}{u^2 + 1} = \frac{1}{6} \text{arctg}\, u + C = \frac{1}{6} \text{arctg} \left(\frac{x+4}{6}\right)$$

*Fazer em casa: $\int \frac{1}{4x^2 + 48x + 128}$